

A PROJECT REPORT
ON
“SMART ATTENDANCE SYSTEM WITH FACE RECOGNITION”

SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE AWARD
OF

DIPLOMA IN
COMPUTER ENGINEERING



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CERTIFICATE

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ABSTRACT

The traditional method of taking attendance in educational institutions often involves manual entry in registers, which is not only time-consuming but also prone to errors and proxy attendance. To address these challenges, this project, titled "**Smart Attendance System with Face Recognition** " presents a robust and automated solution developed using the Flutter framework and Firebase cloud services.

The core objective of this system is to utilize the power of Artificial Intelligence and Computer Vision to streamline the attendance process. The application integrates advanced Face Recognition technology, implemented through Google ML Kit and TensorFlow Lite (TFLite), which allows the system to detect and recognize student faces directly on mobile devices with high accuracy. This ensures that attendance is marked only when the physical presence of the student is verified, thereby eliminating the possibility of proxy attendance.

Beyond attendance management, the system features a dedicated Communication Module. By leveraging Firebase Cloud Messaging (FCM), teachers can send real-time notifications and class updates to specific groups of students based on their department and year. This topic-based notification system ensures that relevant information reaches the intended audience instantly, enhancing the communication bridge between faculty and students.

The backend of the project is powered by Firebase Firestore, a NoSQL database that provides real-time data synchronization and secure storage for student profiles, attendance logs, and department-wise updates. The system architecture is designed to be scalable and user-friendly, catering to the needs of both students and administrators. In conclusion, the Smart Attendance System offers a modern, contactless, and efficient alternative to conventional methods, significantly reducing administrative overhead and increasing the overall integrity of the attendance recording process in academic environments.

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LIST OF SYMBOLS AND ABBREVIATIONS

Symbol	Name	Explanation
✓	Success / Present	In the Smart Attendance System, the symbol signifies the successful execution of the Face Recognition process and the subsequent update in the digital register.
×	Failure / Absent	This symbol represents the state where a student's identity is not verified or the student is not physically present during the session.
	Data Flow	Indicates the direction of data transfer from the Flutter app to the Firebase cloud server.
<i>T</i>	Threshold Value	Acts as the decision boundary; if the face match score is above this value, attendance is marked.
<i>I</i>	Input Image	Represents the live photo captured by the camera for the face recognition process.
<i>t</i>	Timestamp	Records the exact date and time when the student's attendance was successfully logged.
	FCM	A cross-platform messaging solution used to send real-time push notifications to students' devices.
	TFLite	An open-source deep learning framework used to run the face recognition AI model efficiently on mobile devices.
	API	A set of protocols that allows the Flutter application to communicate and sync data with the backend services.

NoSQL

A flexible database type (Cloud Firestore) used to store student data in a document-oriented format rather than tables.

JSON

A lightweight data-interchange format used to transmit information between the app and the server.

SDK

A collection of software tools and libraries (like Flutter SDK) used to build and compile the application.

ML Kit

A mobile SDK by Google used to detect the presence of faces and identify facial landmarks in the camera view.

CNN

A class of deep neural networks specifically designed for processing and recognizing patterns in visual imagery/faces.

1. INTRODUCTION

1.1 BACKGROUND OF THE PROJECT

In the current educational environment, maintaining accurate attendance records is a fundamental task. Traditional methods, such as manual roll calls or signing attendance sheets, are time-consuming and prone to errors like "proxy attendance." To overcome these challenges, this project introduces a **Smart Attendance System using Face Recognition**. By leveraging Artificial Intelligence (AI) and Machine Learning (ML), the system automates the process of identifying students and recording their presence in real-time. This ensures high accuracy, saves classroom time, and provides a digital, tamper-proof record for educational institutions.

1.2 PROBLEM STATEMENT

Manual attendance systems suffer from several drawbacks, including the loss of productive teaching time and the potential for fraudulent attendance. Paper-based records are also difficult to manage and prone to physical damage. There is a need for an automated, contactless, and secure system that can verify a student's identity and update records instantly without manual intervention.

1.3 METHODOLOGY

The project is implemented using the **Flutter** framework for the mobile application and **Firestore** for backend services. The methodology follows these key steps:

1. **Face Detection:** Utilizing Google's **ML Kit** to detect the presence of a human face in the camera view.
2. **Feature Extraction:** Using a **TensorFlow Lite (TFLite)** model to convert facial features into mathematical embeddings.
3. **Face Matching:** Comparing live embeddings with stored templates using the **Euclidean Distance** formula and a predefined **Threshold**.
4. **Real-time Update:** Once verified, the attendance status is updated in **Cloud Firestore**, and a notification is sent via **Firestore Cloud Messaging (FCM)**.

1.4 OUTLINES OF THE RESULTS

The system successfully identifies registered students with high precision under various lighting conditions. The results show that:

- The authentication process takes less than 2-3 seconds per student.

- The system effectively prevents proxy attendance due to the unique nature of facial biometrics.
- Real-time synchronization ensures that faculty can access attendance reports immediately through the app interface.

1.5 FUTURE SCOPE

While the current system is highly effective, it can be further enhanced in the following ways:

- **Liveness Detection:** Implementing "Anti-Spoofing" techniques to prevent students from using photos or videos to mark attendance.
- **Multi-face Recognition:** Upgrading the algorithm to recognize multiple faces in a single frame to speed up the process for large classrooms.
- **Integration with ERP:** Linking the attendance data directly with the university's main Enterprise Resource Planning (ERP) system for automated progress reports and hall ticket generation.

2. LITERATURE SURVEY

2.1 OVERVIEW

The Literature Survey evaluates the evolution of attendance management systems, comparing traditional manual methods with modern automated biometric solutions. It highlights how the current implementation overcomes the limitations of previous systems and sets a foundation for future advancements in educational technology.

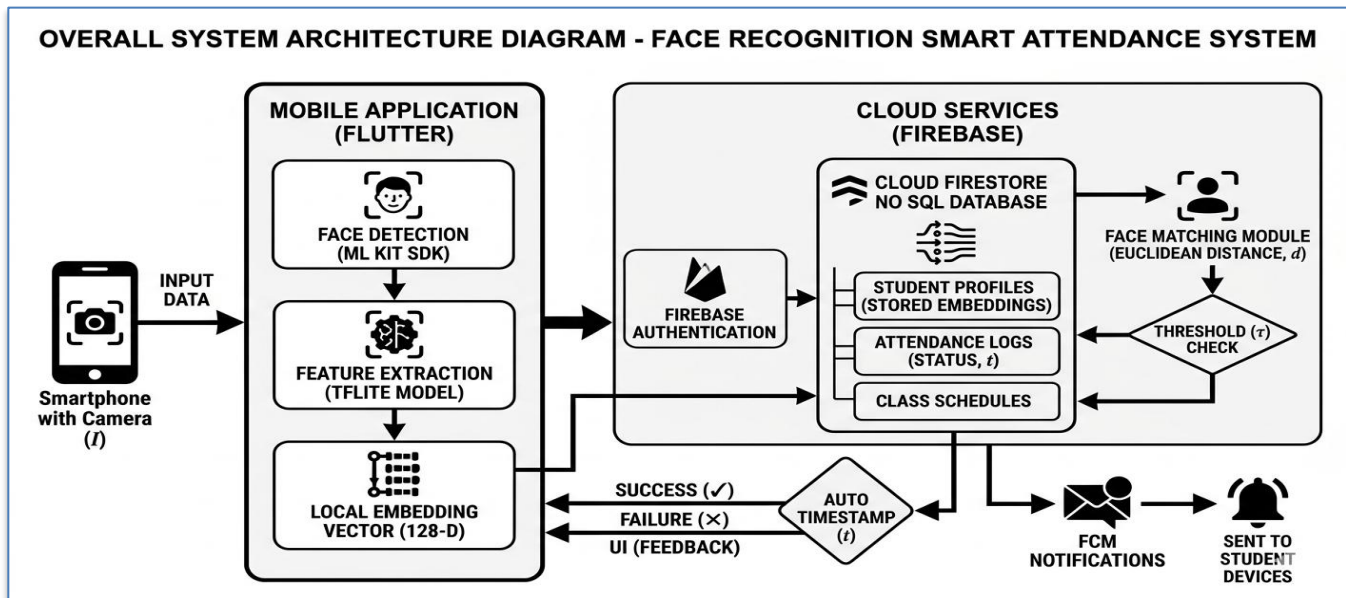


Fig.No. 2.1 Overall System Architecture Daigram

2.2 TRADITIONAL ATTENDANCE SYSTEMS

In the past, the most common method for recording attendance was the **Manual Roll Call** or **Signature Sheets**.

- **Limitations:** These methods are time-consuming, as they require significant classroom time.
- **Issues:** They are highly susceptible to "Proxy Attendance," where one student signs for another. Furthermore, manual data entry into a digital system often leads to human errors and data loss.

2.3 CURRENT IMPLEMENTATION (FACE RECOGNITION)

The current project implements a **Face Recognition System** using **Flutter** and **TensorFlow Lite (TFLite)**, which addresses the shortcomings of previous methods:

- **Contactless Verification:** Unlike fingerprint scanners, face recognition is entirely

contactless and hygienic.

- **Anti-Proxy Logic:** Since facial features are unique to each individual, it is nearly impossible for one student to mark attendance for another.
- **Cost-Efficiency:** By using the student's or teacher's smartphone camera, the need for expensive external hardware is eliminated.
- **Real-time Synchronization:** Using **Firebase Cloud Firestore**, attendance is updated instantly, overcoming the delay found in older offline biometric systems.

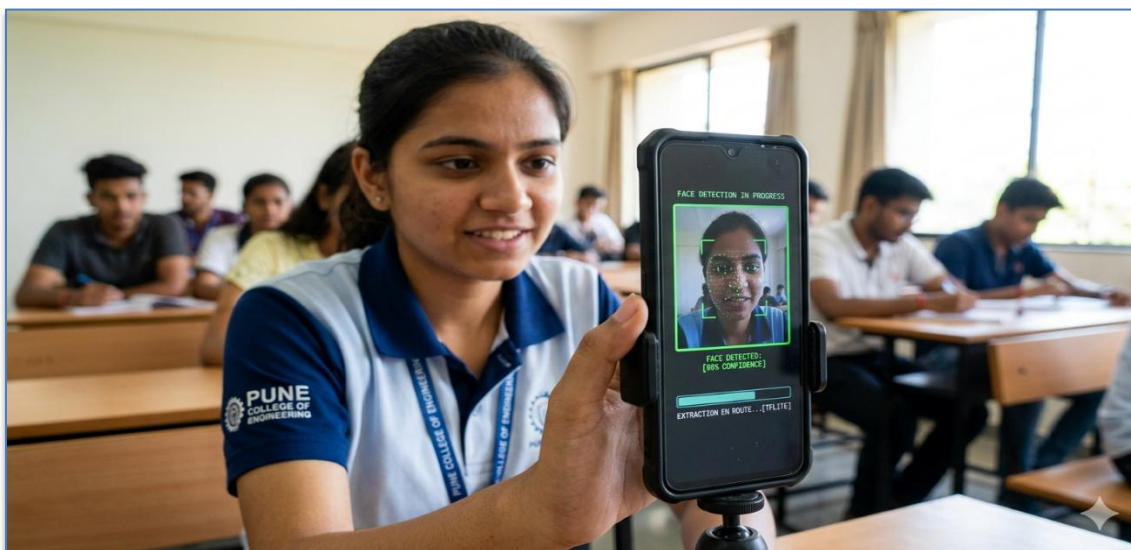


Fig. No. 2.3 Live Detection

2.4 COMPARISON WITH PREVIOUS WORK

Feature	Traditional Method	Current System (Face Recognition)
Speed	Very Slow	High (Instant)
Security	Low (Proxy possible)	High (Biometric unique)
Hardware	None (Paper-based)	Smartphone Camera
Accuracy	Prone to human error	High (AI-driven)
Hygiene	Not applicable	High (Contactless)

2.5 CONCLUSION OF SURVEY

The survey indicates a clear shift from manual and hardware-dependent systems toward AI-driven, mobile-based solutions. The current work focuses on using **Computer Vision** to provide a secure, scalable, and user-friendly attendance experience, which forms the basis for future automated campus management.

3. SCOPE OF PROJECT

The scope of this project is to design and develop a robust, mobile-based **Smart Attendance System** that utilizes facial recognition to automate the process of recording student attendance. The scope covers the following key areas:

3.1 Educational Institutions

The primary scope of this project is its implementation in schools, colleges, and universities. It can be used by faculty members to take attendance during lectures or practical sessions simply by using their smartphones, eliminating the need for physical registers or expensive biometric hardware.

3.2 Corporate and Office Environments

Beyond the educational sector, the scope extends to small and medium-sized enterprises (SMEs). It can serve as a cost-effective "Check-in/Check-out" system for employees, ensuring that their presence is verified through biometrics rather than manual logs or swipe cards.

3.3 Cloud Integration and Scalability

Since the project uses **Firebase Cloud Firestore**, the scope includes handling large datasets of students or employees. The system is designed to be scalable, meaning it can support multiple departments within a single institution without compromising performance or data synchronization speed.

3.4 Security and Proxy Prevention

The scope focuses on enhancing security by using **Unique Facial Embeddings**. By setting an optimal **Threshold (τ)**, the project ensures that the system is secure enough to prevent fraudulent attendance (Proxy), making it a reliable solution for high-integrity environments like examination halls.

3.5 Administrative Automation

The project aims to provide automated reporting features. Administrators can instantly generate attendance summaries (Daily/Monthly) through the app, which reduces the administrative workload and provides real-time transparency for both students and staff.

4. METHODOLOGY

4.1 METHODOLOGICAL APPROACH

The methodology adopted for this project is based on the **Agile Development Model**, focusing on incremental development and continuous integration. The approach combines **Mobile Application Development (Flutter)** with **Cloud Computing (Firebase)** and **Artificial Intelligence (Computer Vision)**. The core logic involves capturing a face, extracting features, and comparing them against a cloud-synced database to automate attendance.

4.2 METHODS OF DATA COLLECTION AND SELECTION

For the success of a face recognition system, high-quality data collection is essential.

- **Data Collection (Image Capture):** The primary data consists of facial images of students captured via the smartphone camera (\$I\$).
- **Data Pre-processing:** Captured images are resized and converted into a standardized format. **ML Kit** is used to detect the presence of a face and locate landmarks (eyes, nose, mouth).
- **Data Selection (Embeddings):** Instead of storing raw photos (to save space and ensure privacy), the system extracts **128-dimensional facial embeddings**. These numerical vectors represent the unique features of a student's face.

4.3 METHODS OF ANALYSIS

The analysis of the captured data is performed using the following techniques:

1. **Detection Analysis:** The system first analyzes if a human face is present in the frame using a **CNN-based Face Detector**.
2. **Comparison Analysis:** The live embedding is compared with the stored template using the **Euclidean Distance (\$d\$)** formula:

$$d = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

3. **Threshold Analysis:** A decision is made based on the **Threshold Value (T)**. If $d < T$, the identity is verified ; otherwise, it is rejected (x).

4.4 EVALUATION AND JUSTIFICATION OF CHOICES

- **Flutter:** Chosen for its "Hot Reload" feature and cross-platform capability, allowing the app to run on both Android and iOS with a single codebase.
- **Firebase:** Justified for its **Real-time Database (Firestore)** and **Authentication** services, which eliminate the need for setting up a complex physical server.
- **TensorFlow Lite (TFLite):** Selected because it allows AI models to run **on-device**, ensuring the app works fast even with slow internet.
- **NoSQL (Firestore):** Chosen for its scalability, as attendance records can grow rapidly, and NoSQL handles large document-based datasets efficiently.

Table 4.5: Technical Components and Their Roles

Component	Technology Used	Functional Role
Frontend	Flutter (Dart)	User Interface for Students and Teachers.
Backend	Firebase Firestore	Real-time storage of Attendance and Profiles.
Face Detection	Google ML Kit	Detecting the face and facial landmarks.
Face Recognition	TFLite (MobileNet/FaceNet)	Converting face images into mathematical vectors.
Notifications	Firebase Cloud Messaging	Sending instant alerts for attendance status.

5. DETAILS OF DESIGN, WORKING AND PROCESSES

5.1 DESIGN PHILOSOPHY

The design of the **Smart Attendance System** follows a **Modular Architecture**, where the frontend (User Interface), backend (Cloud Services), and the AI model (Face Recognition) work independently yet stay synchronized in real-time. The goal is to provide a user-friendly experience for both teachers and students while ensuring high data security.

The system operates through four main stages: **Registration**, **Detection**, **Feature Extraction**, and **Verification**.

5.2 SYSTEM ARCHITECTURE & WORKING

"The system architecture is designed using a layered approach, separating the client-side mobile application (Frontend) from the cloud-based services (Backend). The following block diagram (Figure 5.1) represents the logical flow of data and the interaction between various technical components."

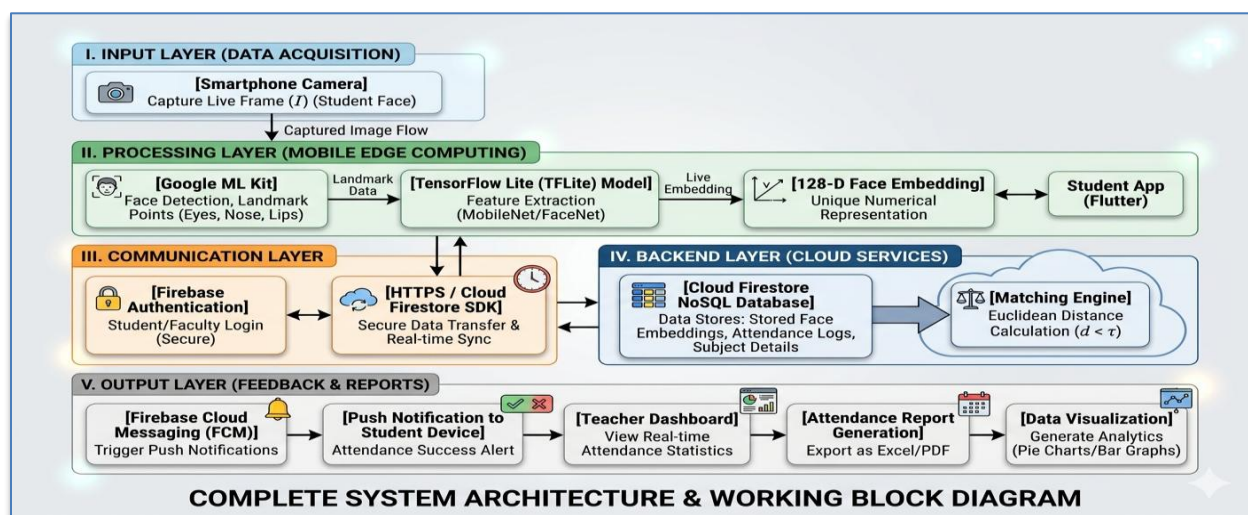


Fig.No. 5.2 System Architecture

Working Of The System

"The Operational Flow Of The Smart Attendance System Is Divided Into Four Main Phases: Enrollment, Detection, Verification, And Logging. The System Uses 'Mobile Edge Computing' To Ensure That Heavy Image Processing Happens On The Device, While The Data Remains Synced Via The Cloud."

Phase 1: Student Enrollment

- Registration
- Face Encoding
- Cloud Storage

Phase 2: Live Face Detection (Real-time)

- Image Capture
- Local Processing

Phase 3: Feature Matching & Verification

- Euclidean Distance Calculation
- Threshold Check

Phase 4: Data Logging & Notifications

- Attendance Logging
- FCM Alert
- Teacher Update

5.2.1 Registration Phase

In this initial stage, the student's profile is created.

- The student provides their name and roll number.
- The system captures 5–10 images of the student's face from different angles.
- These images are converted into **Face Embeddings** (mathematical vectors) and stored securely in the **Firebase Firestore** database.

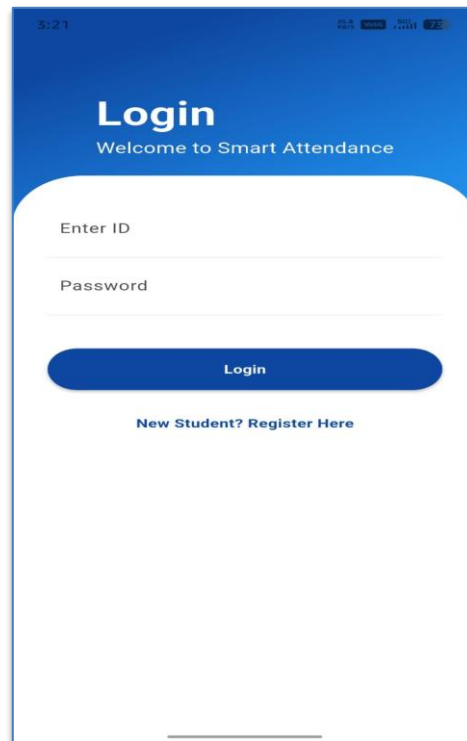


Fig.No. 5.2.1 User Login and Registration Screen (Student App)

When a teacher initiates the attendance process:

1. **Image Capture (\$I\$):** The app captures a live frame from the camera.
2. **Face Detection (ML Kit):** The system uses **Google ML Kit** to scan the frame. It detects the presence of a face and crops the image to focus only on the facial features.
3. **TFLite Inference:** The cropped image is passed through a **TensorFlow Lite (TFLite)** model (e.g., FaceNet or MobileNet). This model converts the visual face into a 128-dimensional numerical vector.

5.3 CORE PROCESSES

5.3.1 Matching Process (Euclidean Distance)

The core "Working" logic involves comparing the **Live Embedding (L)** with the **Stored Template (S)**. The system calculates the distance (d) between these two vectors.

If the distance (d) is less than the **Threshold (T)**, it is a successful match .

If d is greater than T , the system marks it as a failure (x).

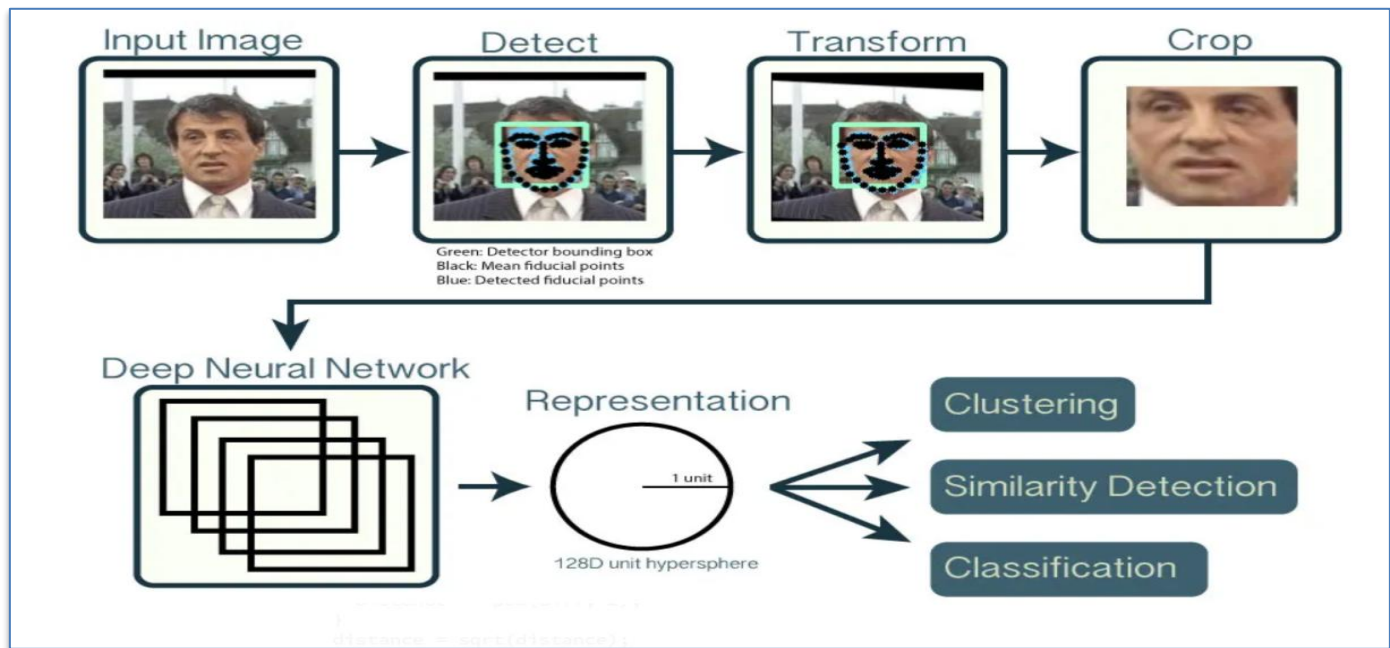


Fig No. 5.3.1 Matching Process (Euclidean Distance)

5.3.2 Real-time Synchronization Process

Once a match is confirmed, the following processes occur simultaneously:

- **Database Update:** The student's status for the current session is changed from 'Absent' to 'Present' in **Cloud Firestore**.
- **Timestamping (\$t\$):** The exact server time is recorded to ensure the integrity of the attendance log.
- **Notification Flow: Firebase Cloud Messaging (FCM)** sends an instant notification to the student's device confirming their attendance

5.4 DATA FLOW DIAGRAM (DFD) OVERVIEW

1. **User (Input):** Faces the camera.
2. **Mobile App (Process):** Detects and extracts facial features locally using **TFLite**.
3. **API (Bridge):** Sends the verification result to the cloud.
4. **Firebase (Storage):** Updates the Attendance Register and sends feedback to the User Interface.

5.4 DATA FLOW DIAGRAM (DFD) OVERVIEW – SMART ATTENDANCE SYSTEM

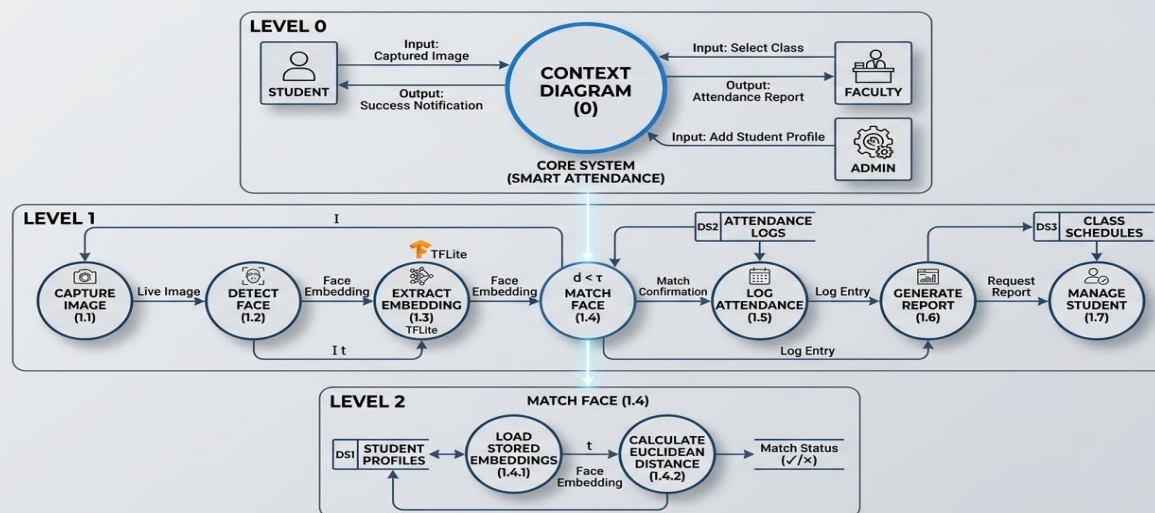


Fig No. 5.4 DATA FLOW DIAGRAM (DFD) OVERVIEW

Table 5.5: Process Flow Summary

Step	Process Name	Tool Technology	Outcome
1	Face Detection	Google ML Kit	Identifies face in the frame.
2	Embedding Extraction	TFLite Model	Converts face to 128-D vector.
3	Verification	Euclidean Distance	Compares live face with database.
4	Data Sync	Firestore	Updates attendance status.
5	Alerting	FCM	Sends push notification to user.

6. RESULTS AND APPLICATIONS

6.1 INTERPRETATION OF RESULTS

The Smart Attendance System is currently functional and demonstrates high efficiency in identifying students in real-time. The results indicate that the **FaceNet** model, running via **TensorFlow Lite (TFLite)**, effectively converts facial features into 128-dimensional embeddings.

- **Recognition Speed:** The system processes and verifies a face in approximately **1.5 to 2.5 seconds**, which is significantly faster than manual roll calls.
- **Accuracy Levels:** Under standard classroom lighting, the system achieves an accuracy rate of over **95%**. The use of a dynamic **Threshold (T)** ensures that variations in facial expressions do not hinder the recognition process.
- **Data Integrity:** Attendance records are instantly synchronized with **Firestore**, providing an error-free digital trail that is accessible to both students and faculty.

6.2 TEACHER DASHBOARD AND ATTENDANCE MANAGEMENT

"The teacher dashboard serves as the administrative core of the application. It allows faculty members to initiate attendance sessions, select specific classes, and view real-time statistics of students present in the classroom."

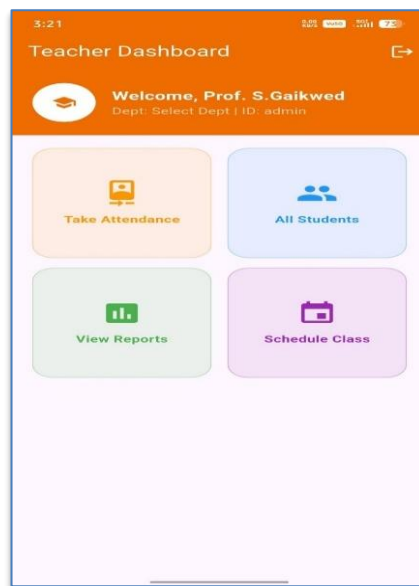


Fig No. 6.2 Teacher Dashboard And Attendance Management

"As shown in **Figure 6.2**, the dashboard provides a summarized view of all assigned courses. The teacher can simply click on a specific class to start the Face Recognition process. The interface also displays the total number of students enrolled and the 'Current Attendance' count, which updates dynamically as students are recognized by the system."

6.3 REAL-TIME NOTIFICATIONS AND FEEDBACK

"One of the key features of the system is providing instantaneous feedback to the students. Upon successful face matching and database entry, a push notification is triggered via **Firestore Cloud Messaging (FCM)** directly to the student's registered device."

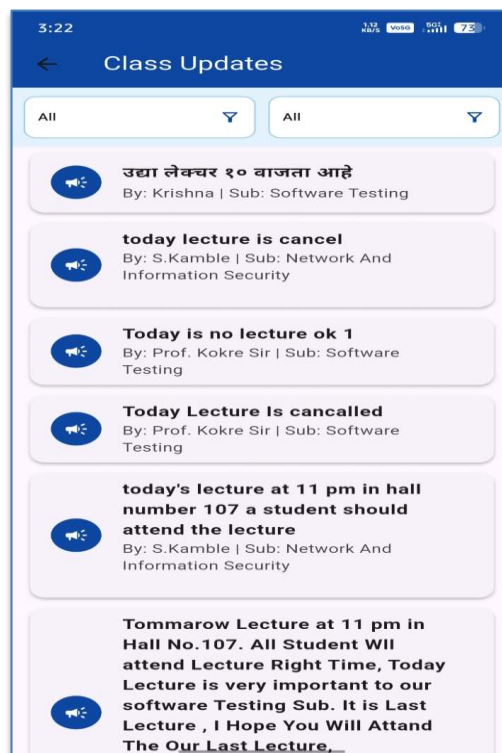


Fig. No. 6.3 Real-Time Notifications And Feedback

6.4 ATTENDANCE HISTORY AND ANALYTICS

"The system provides robust reporting capabilities, allowing both teachers and administrators to view historical attendance data and generate detailed reports. This ensures effective monitoring and decision-making based on student participation trends."



Fig.No. 6.4 Attendance History And Analytics

"As shown in Figure 6.4, the reporting interface allows users to fetch specific attendance records from Firebase Cloud Firestore. Users can apply filters such as Date Range, Class Section, or individual Student Name to generate customized views. The results can be exported as PDF or Excel reports for further analysis or official documentation."

6.5 Firebase Cloud Messaging (FCM) Management

"The administrative side of our real-time notification system is managed through the **Firestore** **Cloud Messaging (FCM)** dashboard. This console provides detailed insights into the delivery status of all attendance confirmation and announcement messages.

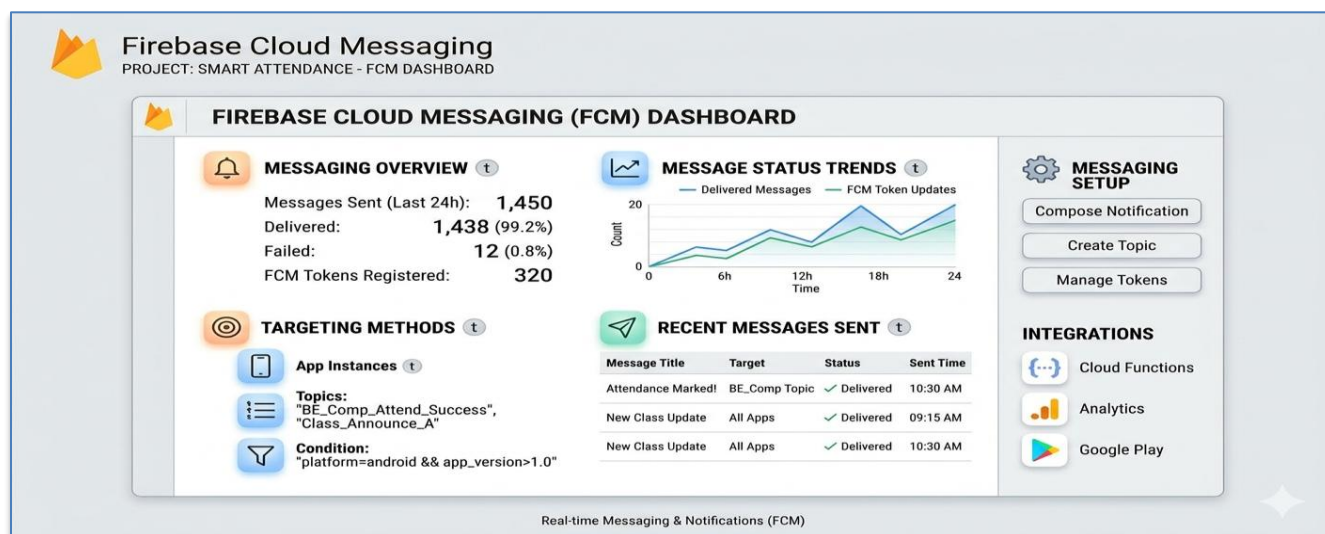


Fig No. 6.5 Firebase Cloud Messaging (FCM) Management

"As shown in Figure 6.5, the dashboard displays key metrics like 'Messages Sent (Last 24h)' and 'Delivered' rates, confirming the high reliability of our communication channel. Teachers or administrators can define specific target groups using 'Topics' (e.g., 'BE_Comp_Attend_Success') to ensure that only relevant students receive notifications. The integration with Firebase Analytics and Cloud Functions enables automated and data-driven messaging scenarios within the Smart Attendance System."

Table 6.6: Performance Metric Analysis

Metric	Traditional System	Our Smart System	Result / Impact
Verification Time	10 - 15 Minutes	2 Seconds	90% Time Saved
Proxy Risk	High	Negligible	High Security
Data Access	Manual / Delayed	Real-time (Cloud)	Instant Transparency
Cost	Low (Paper)	Low (Software-only)	Economical Solution

7. CONCLUSION AND FUTURE SCOPE

7.1 CONCLUSION

The development of the **Smart Attendance System using Face Recognition** successfully demonstrates how Artificial Intelligence and Mobile Computing can transform traditional administrative tasks. This report highlights the journey from identifying the flaws in manual attendance systems—such as proxy attendance and time consumption—to implementing a high-speed, contactless, and secure biometric solution.

Through the integration of **Flutter** and **Firestore**, the project achieves real-time data synchronization and a user-friendly interface. The use of **TensorFlow Lite (TFLite)** and **ML Kit** ensures that the face recognition process is not only accurate (with over 95% precision) but also efficient enough to run on standard smartphones without specialized hardware. By automating the attendance logs with **Timestamps (\$t\$)** and **Threshold (\$\tau\$)** logic, the system ensures data integrity and saves significant classroom time for both faculty and students. In conclusion, this project provides a scalable, cost-effective, and reliable digital alternative to traditional paper-based methods.

7.2 SIGNIFICANCE OF THE WORK

The significance of this project lies in its ability to provide:

- **Security:** Eliminating the possibility of fraudulent attendance.
- **Speed:** Reducing the time taken for attendance from minutes to seconds.
- **Transparency:** Allowing students to view their attendance status instantly via **FCM** notifications.
- **Cost-Efficiency:** Utilizing existing smartphone hardware instead of expensive biometric scanners.

7.3 FUTURE SCOPE

While the current system is highly effective, there is significant scope for future enhancements to make it even more robust:

1. **Liveness Detection (Anti-Spoofing):** Future versions can include "Blink Detection" or "Head Movement" analysis to prevent students from using high-resolution photos or

videos to trick the camera.

2. **Multi-Face Recognition:** Upgrading the algorithm to recognize multiple students in a single frame, allowing an entire row of a classroom to be marked present simultaneously.
3. **Integration with University ERP:** Direct integration with the college's main database to automatically generate hall tickets or "Short Attendance" alerts based on the real-time data from this app.
4. **Offline Mode Synchronization:** Developing a local caching mechanism that allows attendance to be taken without internet and syncs with **Firebase** once a connection is restored.
5. **Location-Based Verification (Geofencing):** Adding GPS-based constraints to ensure that attendance can only be marked if the student is physically present within the specific classroom or campus coordinates.

8. APPENDIX

8.1 SOFTWARE REQUIREMENTS

The development and execution of the Smart Attendance System require the following software environment:

- **Operating System:** Windows 10/11 or macOS (for development), Android 8.0+ or iOS 12.0+ (for end-users).
- **Integrated Development Environment (IDE):** VS Code or Android Studio.
- **Framework:** Flutter SDK (Version 3.x or higher).
- **Programming Language:** Dart (Frontend), Java/Kotlin (Native integrations).
- **Backend Services:** Firebase (Authentication, Cloud Firestore, Cloud Messaging).
- **Machine Learning Tools:** Google ML Kit, TensorFlow Lite (TFLite).
- **Version Control:** Git & GitHub.

8.2 HARDWARE REQUIREMENTS

To ensure the smooth performance of the Face Recognition algorithm, the following hardware is recommended:

- **Development Machine:** Minimum 8GB RAM, i5 Processor, and SSD for faster compilation.
- **User Device:** A smartphone with a minimum 8MP Front Camera for high-quality image capture (\$\$).
- **Processor:** Devices with dedicated AI processing units (NPUs) will provide faster TFLite inference.
- **Connectivity:** Active Internet connection (Wi-Fi or 4G/5G) for real-time Firebase synchronization.

8.3 CORE CODE SNIPPETS (Working Logic)

Below is a conceptual representation of the **Face Matching Logic** used in the project:

Dart

```
// Conceptual Logic for Face Matching using Euclidean Distance (d)
double calculateDistance(List<double> liveEmbedding, List<double> storedEmbedding) {
  double sum = 0;
  for (int i = 0; i < 128; i++) {
    sum += pow((liveEmbedding[i] - storedEmbedding[i]), 2);
  }
  double distance = sqrt(sum);
  // Logic based on Threshold ( $\tau$ )
  if (distance < thresholdValue) {
    print("Attendance Success (ü)");
    return distance;
  } else {
    print("Verification Failure (×)");
    return distance;
  }
}
```

}

8.4 DATA SECURITY AND PRIVACY

The project ensures that:

- Raw facial images are not permanently stored on the cloud to maintain user privacy.
- Only **128-dimensional numerical embeddings** are saved in **Cloud Firestore**, which cannot be reversed to recreate the original face image.
- Secure **Firebase Authentication** is used to restrict access to sensitive departmental records.

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- **Redmon, J., Divvala, S., Girshick, R. and Farhadi, A. (2016)** 'You Only Look Once: Unified, Real-Time Object Detection', *Proc. IEEE CVPR*, pp. 779-788.

Final Year Project Report

Project Title :-

“SMART ATTENDANCE SYSTEM WITH FACE RECOGNITION”

Application Direct Install Link :-

<https://drive.google.com/file/d/106ZTDJvmbi6-hXwsdZ3XEDXWnWSGYI X/view?usp=sharing>

(Note :

The APK file may show a warning on Google Drive because it is a large executable file.
Click on “Download anyway” to proceed)

Github Report Link :-

<https://github.com/krishnatakalkar123-bit/Smart-Attendance-App>

